

A Project Report on

Dairy Farm Management

Submitted to

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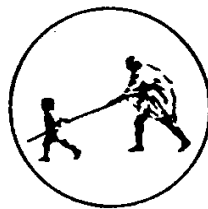
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING
By

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**Under the Guidance
of**

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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)
Academic Year 2025-26**

Certificate



This is to certify that the project entitled

“Dairy Farm Management ”

*Being submitted by **Krushna Jadhav & Rohit Kadam** to the Dr. Babasaheb Ambedkar Technological University, Lonere , for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by him/her under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.*

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ABSTRACT

The Dairy Farm System is a software solution designed to simplify and automate the operations of a Dairy. It manages essential information such as animal records, habitats, feeding schedules, medical treatments, staff details, and visitor management. By replacing traditional manual processes with a centralized digital platform, the system improves data accuracy, enhances coordination, and reduces administrative workload.

The system allows administrators and to efficiently track animal health, dietary needs, and daily activities, while veterinarians can monitor medical histories and schedule timely treatments. Staff duties, ticketing operations, and maintenance activities are also managed systematically to ensure smooth workflow. Through an integrated database and automated modules, the Dairy Management System improves decision-making, enhances animal care, and provides a better experience for both staff and visitors. This technology-driven approach makes zoo administration more efficient, transparent, and reliable.

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With Deep Reverence,

Jadhav Krushana [217]
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CHAPTER 1

INTRODUCTION

The Dairy Farm Shop Management System (DFSMS) is a web-based application that can be accessed over the web. This system can be used to automate the workflow of dairy shop and their invoices. The project has been planned to be having the view of distributed architecture, with centralized storage of the database. The application for the storage of the data has been planned. Using the constructs of MySQL Server and all the user interfaces has been designed using the PHP technologies. The database connectivity is planned using the “MySQL Connection” methodology. The standards of security and data protective mechanism have been given a big choice for proper usage. The application takes care of different modules and their associated reports, which are produced as per the applicable strategies and standards that are put forwarded by the administrative staff. DFSMS is a web-based application which manages the products of dairy shop. It has one module i.e. admin who manages all the functions of the dairy shop. This project is aimed at developing a web based dairy farm shop management System Tool, which is of importance to either a small dairy shop or big dairy shop. The Dairy Shop Management System is a software-based application works as a simple Dairy Software to maintain daily milk record and maintain reports.

The Dairy Farm Management System is a comprehensive web-based application developed to streamline, organize, and automate the day-to-day operations of a dairy farm. Traditionally, dairy farms rely heavily on manual record-keeping methods such as notebooks, registers, or spreadsheets to maintain milk collection data, animal information, and sales records. These manual systems often lead to errors, data loss, difficulty in searching past records, inaccurate calculations, and inefficiency in managing the farm’s overall performance.

To overcome these challenges, the Dairy Farm Management System provides a centralized digital platform where all farm-related information can be stored, processed, and accessed efficiently. The system enables the admin to maintain animal records, milk production data, sales and customer information, and other essential details in an organized manner. It supports quick data entry, secure login, automated calculations, and easy retrieval of historical records. By integrating modern technology with dairy farm operations, the system significantly reduces manual workload, improves accuracy, ensures data security, and enhances operational transparency. It also offers the

ability to generate daily, weekly, or monthly reports, which helps owners analyze production trends, monitor sales, and make better business decisions. With its user-friendly interface and structured database design, the Dairy Farm Management System becomes an essential tool for digitalizing farm management and improving overall productivity.

1.1 Overview

Admin Features:

Dashboard: In this section, admin can see all detail in brief like Total listed categories, companies, products and also see the sales.

Category: In this section, admin can add new categories and edit, delete old categories.

Company: In this section, admin can add new companies and edit, delete old companies.

Product: In this section, admin can add new products and edit old products.

Search: In this section, admin can search for a product then add the product into the cart and generate invoice /receipt.

Invoices: In this section, admin can view all generated invoices/receipts.

Reports: In this section, admin can generate two reports, one is B/w date and another one is for sales.

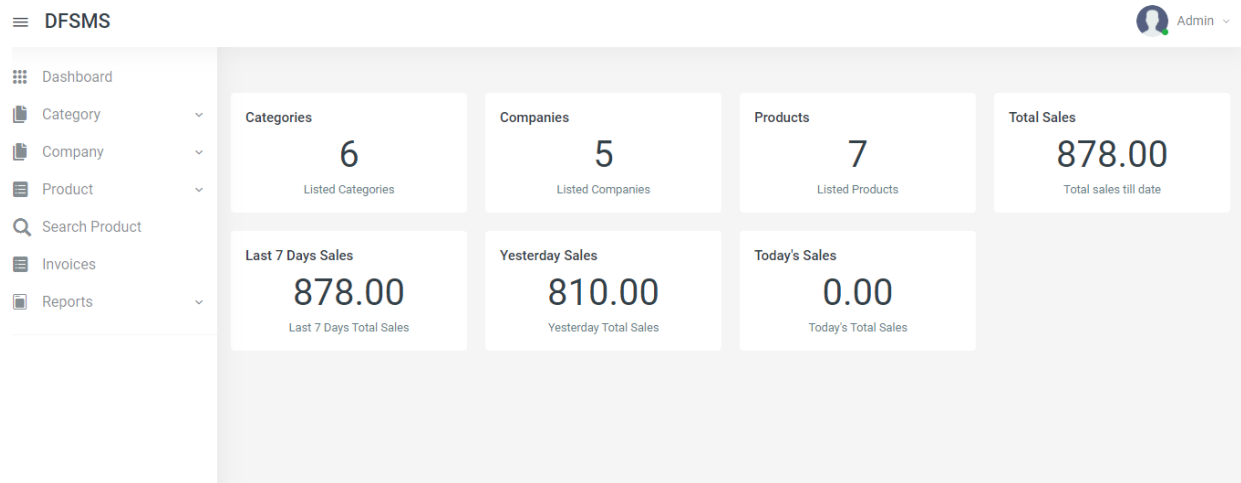
Admin can also update his profile, change the password and recover the password.

1.2 Need for a Dairy Farm Shop Management:

This project is aimed at developing a web based dairy farm shop management System Tool, which is of importance to either a small dairy shop or big dairy shop. The Dairy Shop Management System is a software-based application works as a simple Dairy Software to maintain daily milk record and maintain reports.

- To maintain accurate milk production and selling data
- To track animal health, breed, and maintenance
- To avoid manual errors in calculations

- To generate automated reports
- To increase farm efficiency and profit monitoring



1.3 Problem Statement

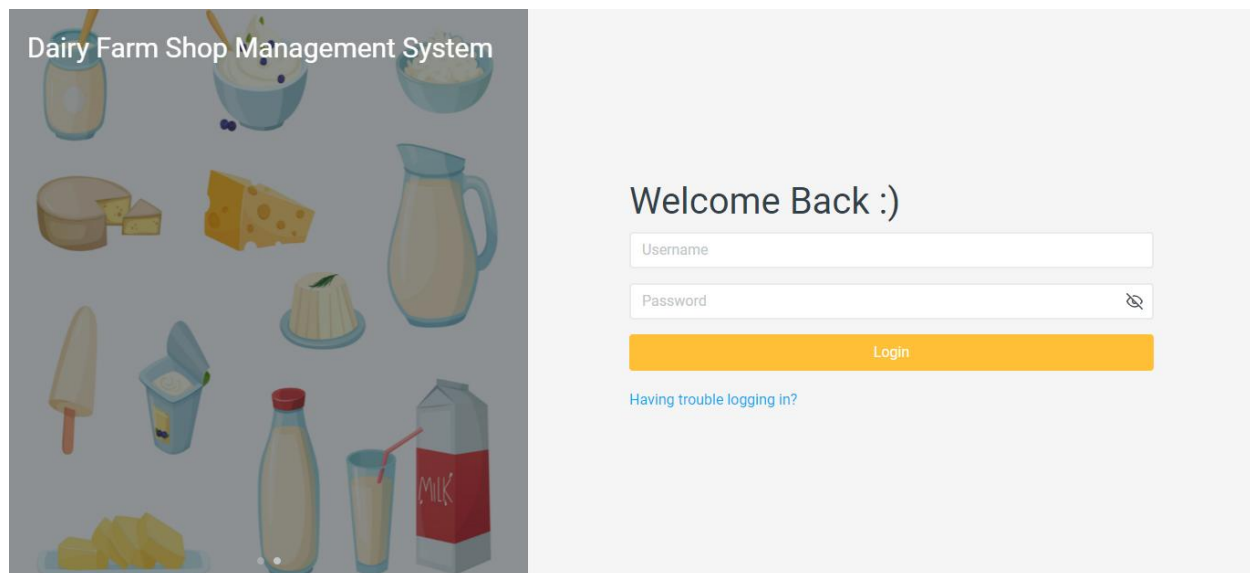
The development of the new system contains the following activities, which try to automate the entire process keeping in view of the database integration approach. Managing the daily operations of a dairy farm using traditional manual methods presents several challenges. Farmers often maintain records of milk production, animal details, and sales transactions in notebooks or registers, which are prone to human error, misplacement, and data inconsistency. As the number of animals and customers increases, it becomes difficult to track milk quantity, maintain animal health histories, update sales information, and retrieve past records accurately. Manual calculations of milk pricing, customer billing, and total production further increase the chances of mistakes and reduce operational efficiency.

- User friendliness is provided in the application with various controls.
- The system makes the overall project management much easier and flexible.
- There is no risk of data mismanagement at any level while the project development is under process.

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1.4 Scope of the Project

The proposed Zoo Management System covers the digital management of animal data, feeding and medical scheduling, staff coordination, and visitor ticketing. It provides functionalities for



adding, updating, monitoring, and retrieving records related to species information, enclosure assignments, healthcare treatments, and nutrition management. It also facilitates the administration of employee roles, duty schedules, and work allocation, while providing systematic recording of visitor details and financial transactions. Additionally, the system offers automated report generation for operational, administrative, and financial analysis to support long-term planning and development

CHAPTER 2

SYSTEM FEASIBILITY ANALYSIS

The Dairy Farm Management System is a web-based application designed to simplify and automate daily dairy farm operations. It helps manage milk collection, animal details, and sales records in a digital and organized way. Software design sits at the technical kernel of the software engineering process and is applied regardless of the development paradigm and area of application. Design is the first step in the development phase for any engineered product or system. The designer's goal is to produce a model or representation of an entity that will later be built. Beginning, once system requirement have been specified and analyzed, system design is the first of the three technical activities -design, code and test that is required to build and verify software.

The importance can be stated with a single word "Quality". Design is the place where quality is fostered in software development. Design provides us with representations of software that can assess for quality. Design is the only way that we can accurately translate a customer's view into a finished software product or system. Software design serves as a foundation for all the software engineering steps that follow. Without a strong design we risk building an unstable system – one that will be difficult to test, one whose quality cannot be assessed until the last stage.

During design, progressive refinement of data structure, program structure, and procedural details are developed reviewed and documented. System design can be viewed from either technical or project management perspective. From the technical point of view, design is comprised of four activities – architectural design, data structure design, interface design and procedural design.

2.1 Management Information System

In most dairy farms, daily activities such as milk collection, animal information, and sales records are maintained manually using notebooks or registers. This traditional system is simple but inefficient as the volume of data increases. Manual record-keeping often leads to errors, missing information, and difficulty in retrieving past records. Calculations for milk quantity, pricing, and sales are done manually, which increases the chances of mistakes. There is no proper system for

DFSMS

Product > Add

Add Product

Category
Select category

Company
Select Company

Product Name
Product Name

Product Price
Product Price

Submit

➤ **Manual Data Entry:**

All milk entries, animal details, and sales information are written by hand, which increases chances of mistakes.

➤ **Data Loss and Damage:**

Registers can be lost, torn, or damaged by water, pests, or mishandling.

➤ **Difficult to Retrieve Information:**

Searching old records manually takes a lot of time and may still lead to incomplete data.

➤ **No Automatic Calculations:**

Pricing, daily totals, fat percentage calculations, and quantity summaries must be done manually, leading to errors.

➤ **Lack of Security:**

Anyone can access or modify the register. There is no password protection or user authentication.

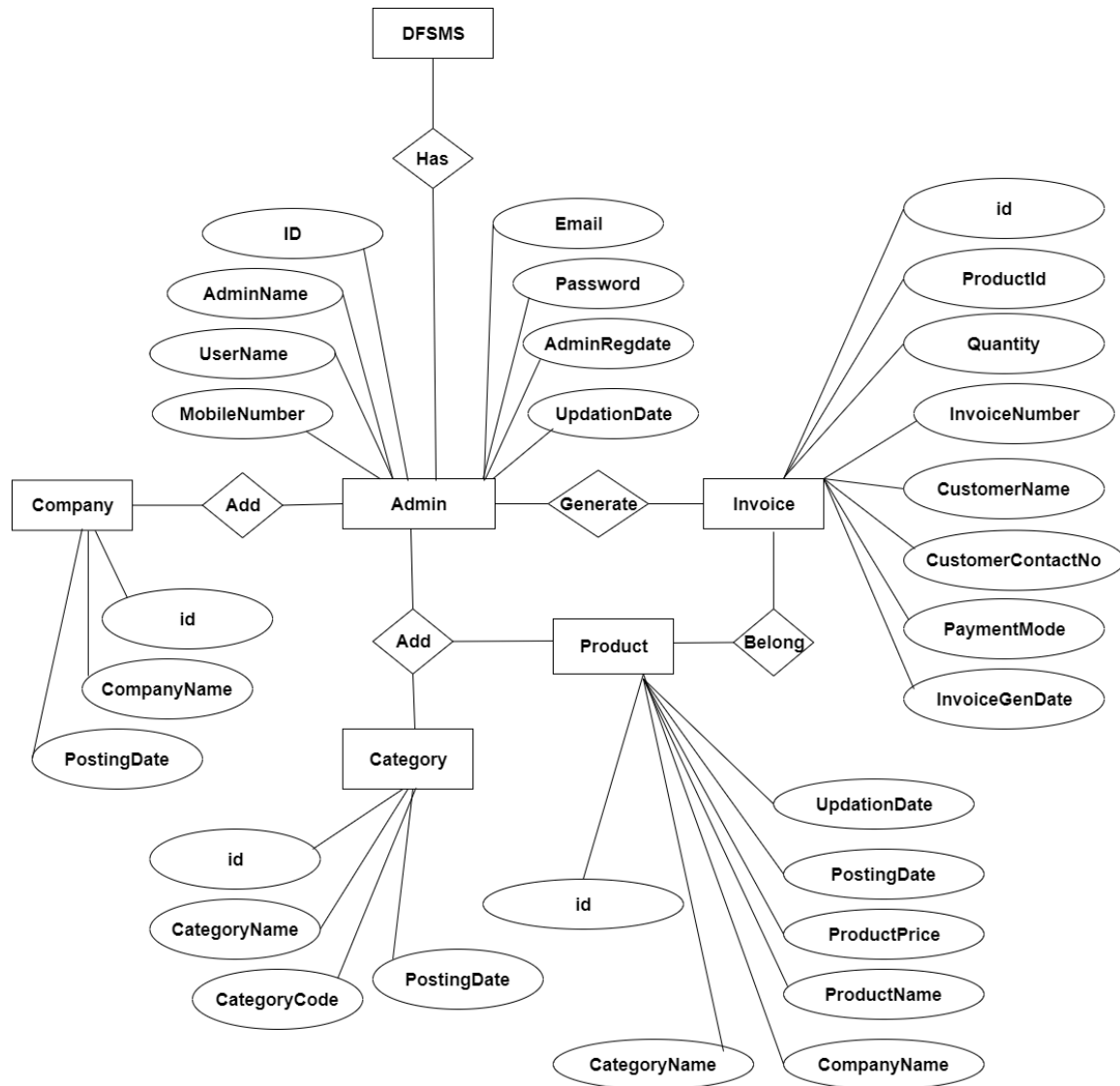
➤ **No Backup:**

If registers are lost or damaged, there is no way to recover the data.

➤ **Low Efficiency:**

Time-consuming processes reduce overall productivity of the dairy farm.

can become unorganized and difficult to manage. The ER diagram helps eliminate such issues by identifying the correct data flow and establishing connections between related modules.



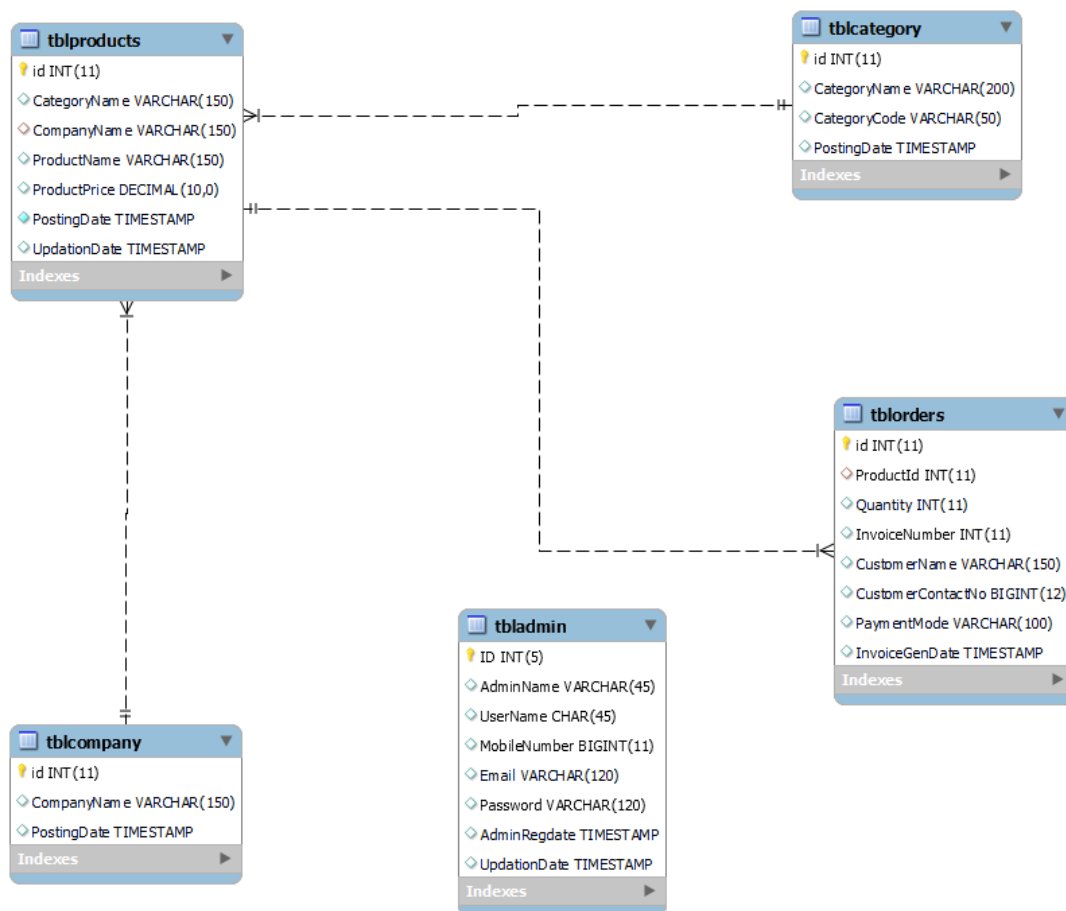
Key Entities Identified:

- **Animal**

Stores details such as animal ID, name, breed, age, and daily milk production.

2.2 System Architecture Overview

The System Architecture of the Dairy Farm Management System is designed to provide a reliable, efficient, and scalable framework for managing all dairy farm operations. The architecture follows a structured three-tier model that separates the user interface, application logic, and data storage layers. This separation ensures smooth communication between components and makes the system easier to maintain and upgrade. The Presentation Layer serves as the front-facing interface where the admin interacts with the system through forms, dashboards, and reports. The Application Layer acts as the core processing unit, executing business rules, validating user inputs, handling authentication, and coordinating all system functions. The Database Layer manages the storage and retrieval of essential data such as milk records, animal details, sales transactions, and user information using a relational MySQL database. This layered architecture not only improves performance and security but also enables modular development, making the system adaptable for future enhancements.



1. Presentation Layer (Frontend)

- Built using HTML, CSS, and JavaScript
- Provides the user interface for the admin
- Handles input forms for milk entry, animal details, and sales
- Displays output such as reports, dashboards, and summaries
- Ensures easy navigation and user-friendly interaction

2. Application Layer (Backend Logic)

- Developed using PHP
- Processes user requests and communicates with the database
- Implements validation, authentication, and business logic
- Performs CRUD operations (Create, Read, Update, Delete)
- Manages modules like Animal Management, Milk Records, Sales, and Reporting

3. Database Layer (Storage Layer)

- Uses MySQL relational database
- Stores all system data: animals, milk entries, customers, sales, admin login
- Maintains data integrity using primary and foreign keys
- Supports fast retrieval through optimized SQL queries
- Ensures data security and backup capability

Overall Workflow

1. The user interacts with the system through the frontend interface.
2. The frontend sends the request to the PHP backend, which processes it.
3. The backend communicates with the MySQL database to store or retrieve data.
4. The result is returned to the frontend, where it is displayed to the user.

SYSTEM ARCHITECTURE OVERVIEW

The System Architecture of the Dairy Farm Management System is built on a structured and efficient three-tier model that separates the system into the user interface, application logic, and database layers. This layered approach ensures that each part of the system performs its own specialized function while maintaining smooth communication with other layers. The user interface layer is responsible for handling all interactions with the admin, providing easy-to-use screens for data entry, updating records, viewing reports, and navigating through the system. The application layer, developed using PHP, manages all business logic such as data validation, processing of milk entries, sales calculations, and overall system workflows. Finally, the database layer, powered by MySQL, securely stores all essential information including animal details, daily milk records, customer profiles, sales transactions, and admin login credentials. By separating these responsibilities, the architecture improves system performance, enhances security, supports scalability, and allows easier future upgrades or module expansion. This structured design ensures that the system remains reliable, maintainable, and capable of meeting the operational demands of a modern dairy farm. By separating concerns across independent layers, the system achieves higher performance, enhanced data

User Interface Layer

- Built using HTML, CSS, JavaScript
- Handles user interactions such as login, data entry, and viewing reports

2. Application Logic Layer

- Implemented in PHP
- Processes all business logic like data validation, calculations, and module control

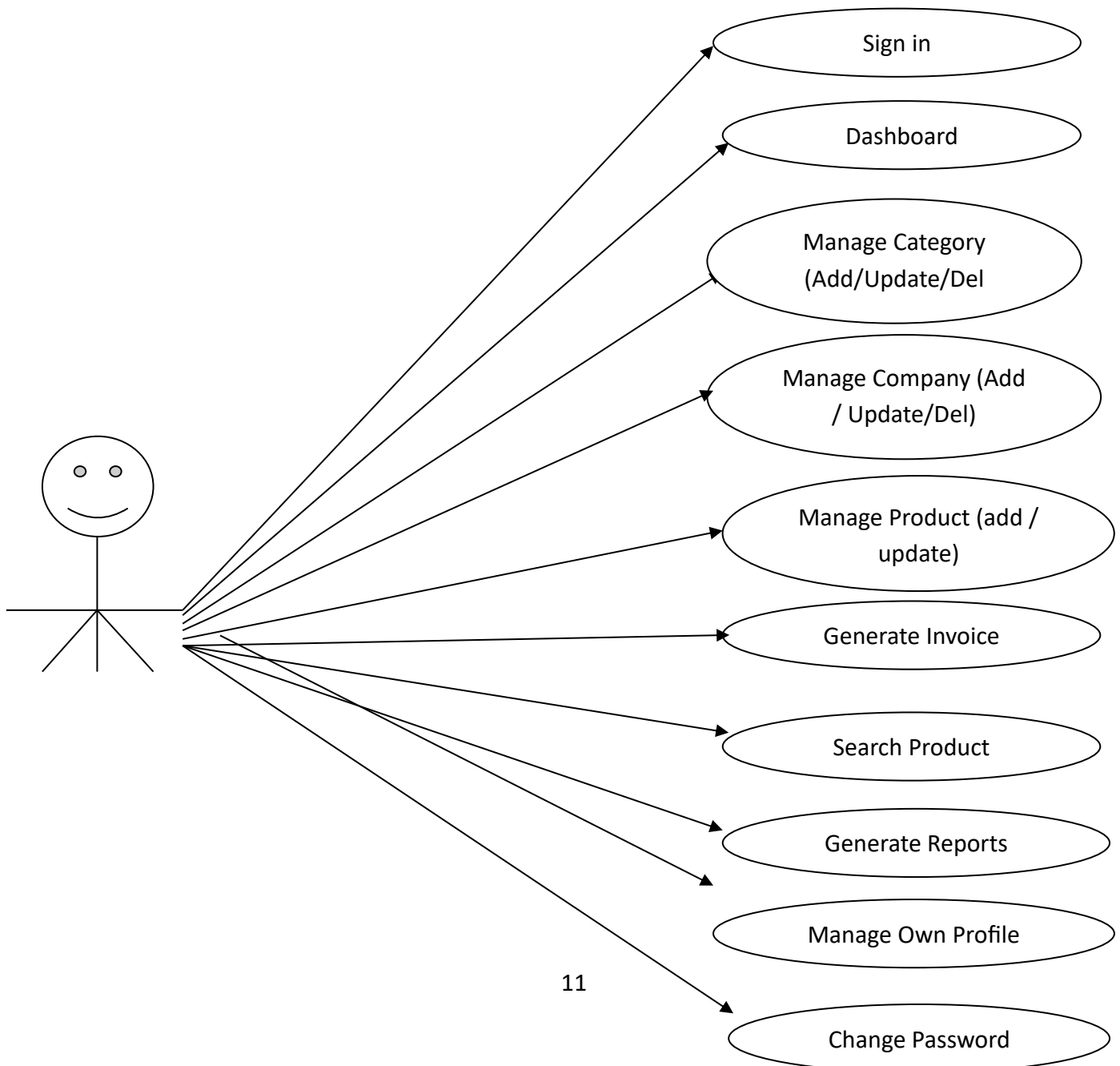
3. Data Storage Layer

- Based on MySQL database

3.1 Use Case Diagram

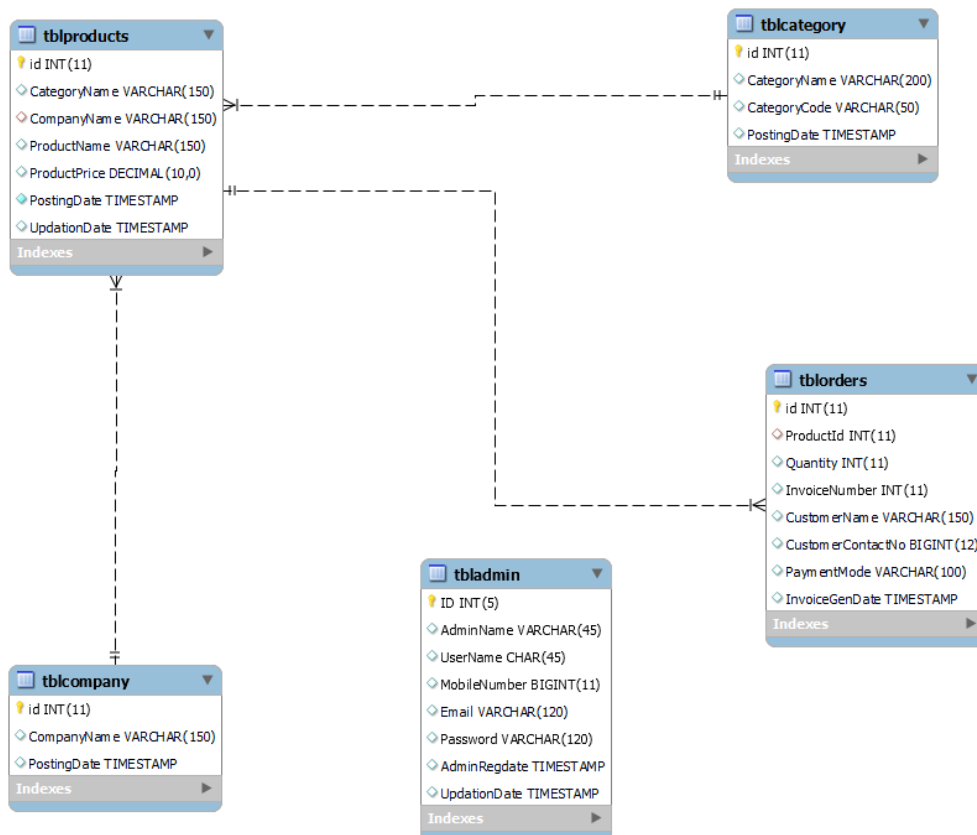
A Use Case Diagram represents the functional requirements of the Dairy Farm Management System by showing how the primary user (Admin) interacts with different system modules. It highlights the major operations such as managing animals, entering milk records, handling sales, generating reports, and logging into the system. This diagram provides a high-level view of the system's behavior, helps identify all user actions, and clarifies the boundaries of the system during the design phase

Use Case Diagram



3.2 Class Diagram

A Class Diagram represents the structural design of the Dairy Farm Management System by defining its main classes, their attributes, methods, and the relationships between them. It provides a blueprint of how data is organized and how different modules—such as Animal, MilkRecord, Customer, Sales, and Admin—interact within the system. This diagram helps developers understand the object-oriented structure, supports database design, and guides the implementation of system functionalities.



This diagram defines how modules are structured and how they connect through relationships such as one-to-many or association.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) illustrates how information moves through the Dairy Farm Management System by showing the flow of data between processes, data stores, and external entities.

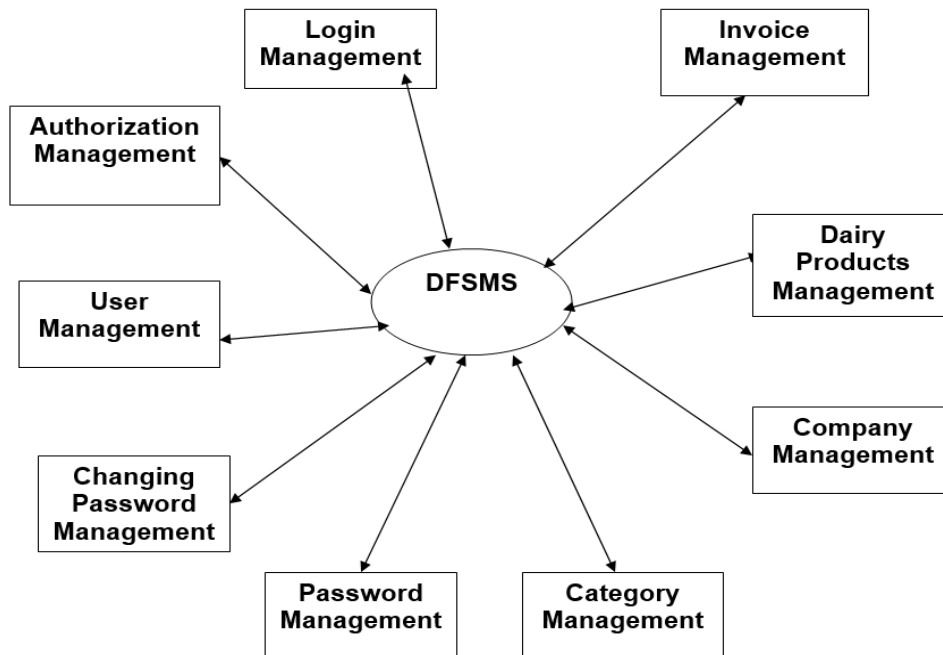
DFD Level 0

Shows the overall system as one process with inputs and outputs from the admin.

DFD Level 1

Breaks the system into smaller processes:

- Animal management
- Milk record processing
- Sales and customer management
- **Zero Level DFD**



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